Informatics College

Module Code & Module Title

MA4001NP Logic and Problem Solving

Assessment Weightage & Type

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Declaration:

I confirm that I understand my coursework needs to be submitted online via My Second Teacher platform under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submission will be treated as non-submission and a mark of zero will be awarded.

Acknowledgment

This project was completed as part of the BSc (Hons) in Computing module MA4001NI – Logic and Problem Solving at Informatics College Pokhara. We would like to express our sincere gratitude to our module leader for providing us with this valuable opportunity to enhance our problem-solving and analytical skills through this coursework.

We are also thankful to all our group members for their cooperation, support, and dedication throughout the completion of this project. Their teamwork, active participation, and continuous effort played a vital role in successfully completing the coursework within the given deadline.

During the preparation of this project, we gained practical knowledge and confidence in several important concepts such as tax calculation systems, linear programming techniques, graphical solutions, simplex methods, break-even analysis, profit maximization, and Excel Solver applications. This coursework also helped us improve our report writing, teamwork, and presentation skills.

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Problem 1 : Tax and Tax Calculation

Taxation is the money that individuals and businesses must pay to the government based on their income. In Nepal, the fiscal year 2025/2026 uses a progressive tax system, which means people who earn more income pay a higher percentage of tax. The government uses this tax money to provide public services like roads, healthcare, education, and other development activities.The tax system is divided into different income slabs, and each slab has its own tax rate. Calculating tax can become difficult when deductions such as Employees Provident Fund (EPF), Life Insurance, and Citizen Investment Trust (CIT) contributions are reduced from the total incomeTo make the process easier, a tax calculation system has been created using Microsoft Excel. This system automatically calculates taxable income, applies the correct tax rates, and provides a detailed report showing yearly tax, monthly tax, total deductions, and the remaining income after tax.
This helps users calculate tax quickly, easily, and accurately.

Term use in tax calculation:

The common terms that I used in tax calculation are explained below in simple language:

Total Yearly Income:

It is the total amount of money a person earns in one fiscal year before any tax or deductions are applied. It includes salary, bonuses, allowances, and other sources of income.

Deductions:

Deductions are the amounts that can be subtracted from total income to reduce taxable income. In this case, deductions include EPF contributions, Life Insurance premiums, and CIT contributions.

Total Yearly Taxable Amount:

This is the income left after deducting EPF, Life Insurance, and CIT from the total yearly income.

The formula is:

Taxable Amount = Total Income − EPF − Life Insurance − CIT

Income Tax Bracket:

An income tax bracket is a range of income that is taxed at a certain rate. People with higher income fall into higher tax brackets.

Income Range	
0	500000
500000	700000
700000	1000000
1000000	2000000
2000000	5000000
5000000	above

Tax Slab:

A tax slab means different parts of income are taxed at different rates. For example, one portion of income may be taxed at 1% and another portion at 10%.

Tax Rate: The tax rate is the percentage used to calculate tax on a certain portion of taxable income.

Tax Rate
1%
10%
20%
30%
36%
39%

Taxable Amount Left:

It is the remaining income after tax has been calculated for one slab. The remaining amount is then moved to the next slab for further tax calculation.

Yearly Tax and Monthly Tax:

Yearly tax is the total tax paid in one year, while monthly tax is the tax amount paid each month.

The formula is:

$$\text{Monthly Tax} = \frac{\text{Yearly Tax}}{12}$$

Income Left After Tax:

It is the amount of money remaining after all taxes have been deducted from the total yearly income. This is the actual income a person can use or save.

Solving the problem

We have solved this problem by creating a spreadsheet in excel that will solve the tax related problem.

Tax template:

Yearly Income	0
Deductions	
PF	0
CIT	0
Insurance	0
Total Deductions	0
Total Taxable Income	0

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount le
0	500000	500000	1%	0	0
500000	700000	200000	10%	0	0
700000	1000000	300000	20%	0	0
1000000	2000000	1000000	30%	0	0
2000000	5000000	3000000	36%	0	0
5000000	above	0	39%	0	
			Yearly Tax	0	
			Monthly Tax	0	
			Yearly Income after Tax:	0	

Figure 1 : Tax template

Tax Formula template:

Yearly Income	0
Deductions	
PF	0
CIT	0
Insurance	0
Total Deductions	=D6+D7+D8
Total Taxable Income	=D3-D10

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount left
0	500000	=C16-B16	0.01	=IF(D12<=D16,E16*D12,E16*D16)	=IF(D12<=D16,0,D12-D16)
500000	700000	=C17-B17	0.1	=IF(G16<D17,G16*E17,D17*E17)	=IF(G16<=D17,0,G16-D17)
700000	1000000	=C18-B18	0.2	=IF(G17<D18,G17*E18,D18*E18)	=IF(G17<=D18,0,G17-D18)
1000000	2000000	=C19-B19	0.3	=IF(G18<D19,G18*E19,D19*E19)	=IF(G18<=D19,0,G18-D19)
2000000	5000000	=C20-B20	0.36	=IF(G19<D20,G19*E20,D20*E20)	=IF(G19<=D20,0,G19-D20)
5000000	above	=IF(D12>=B21,D12-B21,0)	0.39	=IF(G20<D21,G20*E21,D21*E21)	
			Yearly Tax	=SUM(F16:F21)	
			Monthly Tax	=F23/12	
			Yearly Income after Tax:	=D3-F23	

Figure 2: Tax formula template

Test values

a. Tax (Rs 1,45,05,000)

Deduction:

- Employees provident fund organization: (Rs 10,00,000)
- Life insurance premium: (Rs 680000)
- Citizen Investment Trust: (Rs 150000)

Yearly Income	1,45,05,000
Deductions	
PF	1000000
CIT	150000
Insurance	680000
Total Deductions	1830000
Total Taxable Income	12675000

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount left
0	500000	500000	1%	5000	12175000
500000	700000	200000	10%	20000	11975000
700000	1000000	300000	20%	60000	11675000
1000000	2000000	1000000	30%	300000	10675000
2000000	5000000	3000000	36%	1080000	7675000
5000000	above	7675000	39%	2993250	
			Yearly Tax	4458250	
			Monthly Tax	371520.8333	
			Yearly Income after Tax:	10046750	

Figure 3 : Tax calculation for the income of 1,45,05,000

An annual income of Rs 1,45,05,000 is taxed under six different tax slabs. After deductions of Rs 18,30,000, the taxable income becomes Rs 1,26,75,000. The first Rs 5,00,000 is taxed at 1%, which gives a tax of Rs 5,000. The next Rs 2,00,000 is taxed at 10%, adding Rs 20,000 in tax. Income between Rs 7,00,000 and Rs 10,00,000 is taxed at 20%, resulting in Rs 60,000 tax. The next slab, from Rs 10,00,000 to Rs 20,00,000, is taxed at 30%, which adds Rs 3,00,000 tax. Income between Rs 20,00,000 and Rs 50,00,000 is taxed at 36%, giving Rs 10,80,000 tax. The remaining Rs 76,75,000 above Rs 50,00,000 is taxed at 39%, resulting in Rs 29,93,250 tax. Altogether, the total annual tax paid is Rs 44,58,250, which is about Rs 3,71,520.83 per month. After paying all taxes, the remaining income is Rs 82,16,750 from the taxable income, or Rs 1,00,46,750 when calculated from the original gross income.

b. Tax (Rs 6,80,000)

Deductions:

- Employee provident fund organization: (Rs 10,75,000)
- Life Insurance premium: (Rs 2,50,000)
- Citizen Investment Trust: (RS.0)

Yearly Income	80,09,090
Deductions	
PF	1075000
CIT	0
Insurance	250000
Total Deductions	1325000
Total Taxable Income	6684090

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount left
0	500000	500000	1%	5000	6184090
500000	700000	200000	10%	20000	5984090
700000	1000000	300000	20%	60000	5684090
1000000	2000000	1000000	30%	300000	4684090
2000000	5000000	3000000	36%	1080000	1684090
5000000	above	1684090	39%	656795.1	
			Yearly Tax	2121795.1	
			Monthly Tax	176816.2583	
			Yearly Income after Tax:	5887294.9	

Figure 4 : Tax calculation for the amount of 6,80,000

An annual income of Rs 80,09,090 is taxed under six different tax slabs. After deductions, the taxable income becomes Rs 66,84,090. The first Rs 5,00,000 is taxed at 1%, resulting in Rs 5,000 tax. The next Rs 2,00,000 is taxed at 10%, adding Rs 20,000 tax. Income between Rs 7,00,000 and Rs 10,00,000 is taxed at 20%, which gives Rs 60,000 tax. The next slab, from Rs 10,00,000 to Rs 20,00,000, is taxed at 30%, adding Rs 3,00,000 tax. Income between Rs 20,00,000 and Rs 50,00,000 is taxed at 36%, resulting in Rs 10,80,000 tax. The remaining Rs 16,84,090 above Rs 50,00,000 is taxed at 39%, which gives Rs 6,56,795.10 tax. Altogether, the total annual tax paid is Rs 20,53,295.10, which is about Rs 1,71,107.93 per month. After all taxes are deducted, the remaining income is Rs 59,55,794.90.

c. Tax (Rs. 49,10,000)

Deduction:

- Employees provident fund organization: (Rs 4,44,000)

- Life Insurance premium: (Rs. 350,000)

- Citizen Investment Trust:((Rs.70,000)

Yearly Income	49,10,000
Deductions	
PF	444000
CIT	70000
Insurance	350000
Total Deductions	864000
Total Taxable Income	4046000

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount left
0	500000	500000	1%	5000	3546000
500000	700000	200000	10%	20000	3346000
700000	1000000	300000	20%	60000	3046000
1000000	2000000	1000000	30%	300000	2046000
2000000	5000000	3000000	36%	736560	0
5000000	above	0	39%	0	
			Yearly Tax	1121560	
			Monthly Tax	93463.33333	
			Yearly Income after Tax:	3788440	

Figure 5: Tax calculation for the amount of 49,10,000

An annual income of Rs 49,10,000 is taxed under five different tax slabs. After deductions, the taxable income becomes Rs 40,46,000. The first Rs 5,00,000 is taxed at 1%, resulting in Rs 5,000 tax. The next Rs 2,00,000 is taxed at 10%, adding Rs 20,000 tax. Income between Rs 7,00,000 and Rs 10,00,000 is taxed at 20%, which gives Rs 60,000 tax. The next slab, from Rs 10,00,000 to Rs 20,00,000, is taxed at

30%, adding Rs 3,00,000 tax.

The remaining Rs 20,46,000 is taxed at 36%, resulting in Rs 7,36,560 tax. Altogether, the total annual tax paid is Rs 11,21,560, which is about Rs 93,463.33 per month. After all taxes are deducted, the remaining income is Rs 37,88,440.

d. Tax (Rs 22,08,790)

Deductions:

- Employees provident Fund organization: (Rs 2,50,000)
- Life Insurance premium : (Rs 0)
- Citizen Investment Trust: (Rs.2,05,000)

Yearly Income	22,08,790
Deductions	
PF	250000
CIT	205000
Insurance	0
Total Deductions	455000
Total Taxable Income	1753790

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount left
0	500000	500000	1%	5000	1253790
500000	700000	200000	10%	20000	1053790
700000	1000000	300000	20%	60000	753790
1000000	2000000	1000000	30%	226137	0
2000000	5000000	3000000	36%	0	0
5000000	above	0	39%	0	
			Yearly Tax	311137	
			Monthly Tax	25928.08333	
			Yearly Income after Tax:	1897653	

Figure 6:Tax calculation for the amount of 22,08,790

An annual income of Rs 22,08,790 is taxed under four different tax slabs. After deductions, the taxable income becomes Rs 17,53,790. The first Rs 5,00,000 is taxed at 1%, resulting in Rs 5,000 tax. The next Rs 2,00,000 is taxed at 10%, adding Rs 20,000 tax. Income between Rs 7,00,000 and Rs 10,00,000 is taxed at 20%, which gives Rs 60,000 tax. The remaining Rs 7,53,790 is taxed at 30%, resulting in Rs 2,26,137 tax. Altogether, the total annual tax paid is Rs 3,11,137, which is about Rs 25,928.08 per month. After all taxes are deducted, the remaining income is Rs 18,97,653.

e. Tax (Rs 3,10,800)

Deduction:

- Employees provident Fund organization: (Rs 75,000)

- Life Insurance premium : (Rs 5,000)

- Citizen Investment Trust: (Rs. 20,000)

Yearly Income	3,10,800
Deductions	
PF	75000
CIT	20000
Insurance	5000
Total Deductions	100000
Total Taxable Income	210800

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount left
0	500000	500000	1%	2108	0
500000	700000	200000	10%	0	0
700000	1000000	300000	20%	0	0
1000000	2000000	1000000	30%	0	0
2000000	5000000	3000000	36%	0	0
5000000	above	0	39%	0	
			Yearly Tax	2108	
			Monthly Tax	175.6666667	
			Yearly Income after Tax:	308692	

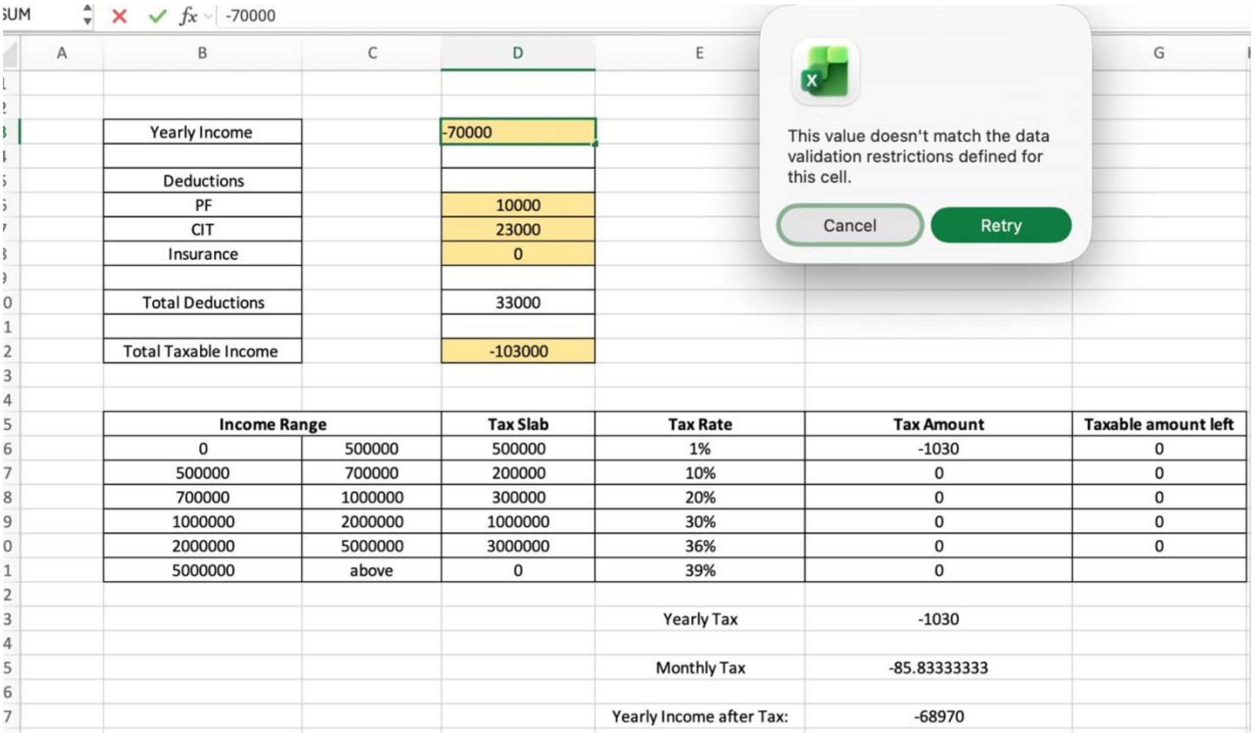
Figure 7:Tax calculation for the amount of 3,10,800

An annual income of Rs 3,10,800 is taxed under the first tax slab only. After deductions, the taxable income becomes Rs 2,10,800. Since this amount falls within the Rs 0 to Rs 5,00,000 slab, a tax rate of 1% is applied, resulting in a tax amount of Rs 2,108. Therefore, the total annual tax paid is Rs 2,108, which is about Rs 175.67 per month. After all taxes are deducted, the remaining income is Rs 3,08,692.

f. Tax (RS -70,000)

Deductions:

- Employee provident fund organization: (Rs10,000)
- Life Insurance premium: (Rs 0)
- Citizen Investment Trust: (Rs 23,000)



Yearly Income	-70000
Deductions	
PF	10000
CIT	23000
Insurance	0
Total Deductions	33000
Total Taxable Income	-103000

Income Range		Tax Slab	Tax Rate	Tax Amount	Taxable amount left
0	500000	500000	1%	-1030	0
500000	700000	200000	10%	0	0
700000	1000000	300000	20%	0	0
1000000	2000000	1000000	30%	0	0
2000000	5000000	3000000	36%	0	0
5000000	above	0	39%	0	

Yearly Tax	-1030
Monthly Tax	-85.83333333
Yearly Income after Tax:	-68970

Figure 8:Tax calculation for the amount of -70,000

This procedure shows an error message whenever a user enters invalid data. It helps users understand what went wrong and how to fix it. In Excel, this process is called data validation, where rules are set to control the type of data users can enter. For example, a yearly income cannot be negative because it is not possible in real life. If a negative value is entered, the system displays the message: “Error: Invalid input. Yearly income cannot be negative. Please enter a valid positive income.” This method helps keep the data accurate, consistent, and reliable by preventing incorrect entries.

Problem 2

Part A :

Prakash Electric Bulb Company has three production plants, each of which produces three different models of an electric bulb. The daily capacities of the three plants are as follows.

Plant / Model	Model 1	Model 2	Model 3
Plant 1	8000	4000	8000
Plant 2	6000	6000	3000
Plant 3	12000	4000	8000

The total demand for Model 1 is 300,000 units, for Model 2 is 172,000 units, and for Model 3 is 249,500 units. Moreover, the daily operating cost for Plant 1 is Rs.55,000, for Plant 2 is Rs.60,000, and for Plant 3 is also Rs.60,000. How many days should each plant be operated in order to fill the total demand, and keep the operating cost at a minimum?

a) Formulate the problem as a linear programming model, clearly defining the variables, the objective function, and the constraints.

Decision Variables :

X_1 = Number of days Plant 1 should be operated
X_2 = Number of days Plant 2 should be operated
X_3 = Number of days Plant 3 should be operated

Objective Function
Minimize : $Z = 55{,}000X_1 + 60{,}000X_2 + 60{,}000X_3$

Table 1 : Objective function

Plant / Model	**Model 1 (units/day)**	**Model 2 (units/day)**	**Model 3 (units/day)**
Plant 1	8,000	4,000	8,000
Plant 2	6,000	6,000	3,000
Plant 3	12,000	4,000	8,000

Constraints :

Model 1 : $8000X_1 + 6000X_2 + 12000X_3 \geq 300{,}000$ units
Model 2 : $4{,}000x_1 + 6{,}000x_2 + 4{,}000x_3 \geq 172{,}000$ units
Model 3 : $8{,}000x_1 + 3{,}000x_2 + 8{,}000x_3 \geq 249{,}500$ units

Solve the problem using Simplex method :

To apply the simplex method, the constraints are converted to standard form ;

Converts $\geq$ constraints to $\leq$ by multiplying by -1.

The inequalities are converted into equations using surplus and artificial variables :

$8000x_1 + 6000x_2 + 12000x_3 - s_1 + a_1 = 300000$

$4000x_1 + 6000x_2 + 4000x_3 - s_2 + a_2 = 172000$

$8000x_1 + 3000x_2 + 8000x_3 - s_3 + a_3 = 249500$

Optimal Solution :

Using the simplex method , the optimal solution is :

Table 2: Optimal Solution

Plant	**Operating Days**
Plant 1	22.5 days
Plant 2	10.5 days
Plant 3	4.75 days

Minimum Operating Cost

Z = 55000(22.5) + 60000(10.5) + 60000(4.75)

Z = 1237500 + 630000 + 285000

Z = Rs. 2,152,500

Verification Of Constraints :

Model	Required(units)	Produced(units)	Status
Model 1	300,000	300,000	Met
Model 2	172,000	172,000	Met
Model 3	249,500	249,500	Met

Cost Breakdown by Plant :

Table 3:Cost breakdown by plant

Plant	**Days**	**Daily Cost**	**Total Cost**	**% of Total**
Plant 1	22.50	Rs. 55,000	Rs. 1,237,500	57.5%
Plant 2	10.50	Rs. 60,000	Rs. 630,000	29.3%
Plant 3	4.75	Rs. 60,000	Rs. 285,000	13.2%
TOTAL	37.75		Rs. 2,152,500	100%

Solving by Excel :

	Label	Value
	22.5	(decision variable)
	10.5	decision variable)
	4.75	decision variable)
MinimumTotal Cost	2152500	(decision variable)
Model 1	300000	
Model 2	172000	
Model 3	249500	

Figure 9:Solving by excel

Memorandum :

To : Parkash Electric Bulb Company

From : Operations Adviser

Subject : Recommended Production Schedule

Date : 21 May 2026

Executive Summary

After analyzing the production and operating cost of all the three plants, the economical production schedule has been determined.

The recommended operating schedule is :

Plant 1 – Operate for 22.5 days per month

Plant 2 – Operate for 10.5 days per month

Plant 3 – Operate for 4.75 days per month

Current Situation

Your company has three plants with different operating costs :

Plant 1 costs Rs. 55000 per day to operate

Plant 2 costs Rs. 60000 per day to operate

Plant 3 – costs Rs. 60000 per day to operate

Each plants produce all three bulb models, but at different rate. The following customer demand should be met :

Model 1: 300,000 units

Model 2: 172,000 units

Model 3: 249,500 units

Key Recommendations :

Plant Operating Schedule

Plant 1 should operate for 22.5 days because it has the lowest daily operating cost.

Plant 2 should operate for 10.5 days to help meet the remaining demand that Plant 1 couldn’t cover.

Plant 3 should operate for 4.75 days to fulfil the final parts of demand.

This plan ensures :

Minimum operating cost

Full demand satisfaction

No overproduction

Efficient use of all plants

Part B : Linear Program Solution

Solve the following linear programming problem graphically.

Maximize $Z = 13x + 11y$

Subject to constraints,

- $4x + 5y \leq 1500$ (Constraint 1)
- $5x + 3y \leq 1575$ (Constraint 2)
- $x + 2y \leq 420$ (Constraint 3)
- $x, y \geq 0$ (Non - negativity)

Graphical Solution

Maximize :

$Z = 13x + 11y$

Subject to:

$4x + 5y \leq 1500$

$5x + 3y \leq 1575$

$x + 2y \leq 420$

$x , y \geq 0$

Constraint Analysis :

Constraint 1 ($4x + 5y = 1,500$): x-intercept = 375, y - intercept = 300

Constraint 2 ($5x + 3y = 1,575$): x-intercept = 315, y - intercept = 525

Constraint 3 ($x + 2y = 420$): x-intercept = 420, y-intercept = 210

Graphical Representation of feasible region :

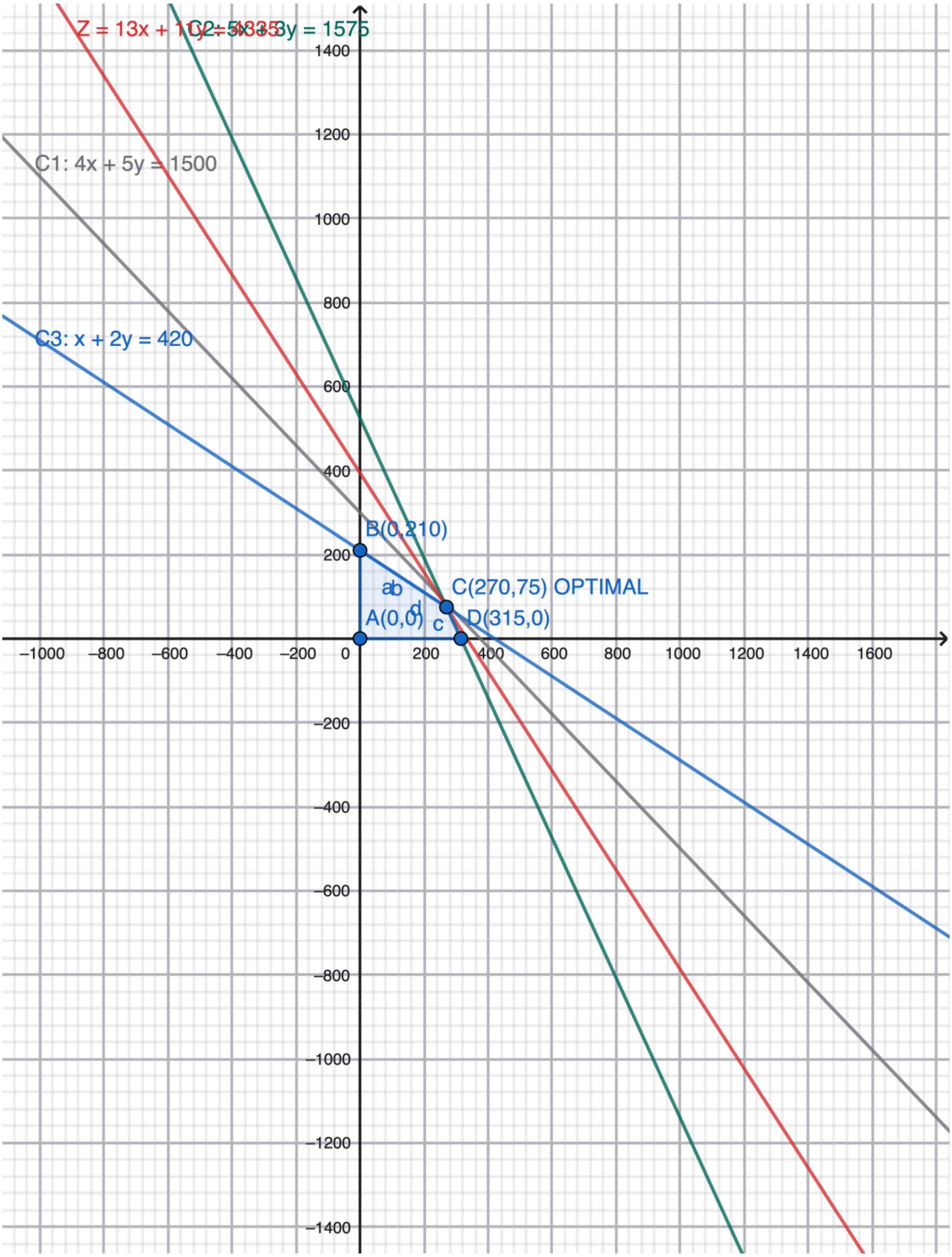


Figure 10:Graphical representation of feasible region

Corner Points Analysis :

Table 4:Corner points analysis

Corner Point	x	y	C1: 4x+5y	C2: 5x+3y	C3 : x+2y	feasible	Z= 13x+11y
(0,0)	0.00	0.00	0 ≤ 1500	0 ≤ 1575	0 ≤ 420	YES	0.00
(0, 210)	0.00	210.00	1050 ≤ 1500	630 ≤ 1575	420 ≤ 420	YES	2,310.00
(270, 75)	270.00	75.00	1455 ≤ 1500	1575 ≤ 1575	420 ≤ 420	YES	4,335.00 0
(315, 0)	315.00	0.00	1260 ≤ 1500	1575 ≤ 1575	315 ≤ 420	YES	4,095.00

Problem 3

In a survey conducted by a research department, Sonam Plastic Industry manufactures a certain type of product. The company's total fixed costs is Rs. 30 and the variable cost per unit is Rs. 4. The total revenue function for the company has been determined by the market research team which is $\mathbf{R = 39x - 5x^2}$, where R represents the total revenue in Rupees and x represents the quantity of the product produced and sold. The sections of your report should correspond to the individual questions below:

(a) Find the total cost function C and calculate the breakeven points.

(b) Plot the revenue and total cost functions on graph, showing the accurate break-even output levels and corresponding price.

(c) Determine the profit function P(x) for the industry and find, (i) The level of production that maximizes the profit.

(ii) The maximum profit. Round the answers to the nearest integer values where applicable.

a. Cost function :

A cost function is a mathematical expression that calculate the total cost of producing a certain quantity of output based on input prices and the number of units produced. It is commonly utilized by companies to analyze production costs and optimize efficiency, typically represented graphically as a cost curve (Moffatt, 2018). The cost function helps businesses understand how their expenses change as production levels increase or decrease.

The cost function equation is generally expressed as: C(x) = FC + VC(x) where,

C(x) =equals total production cost,

FC =total fixed costs

VC(x) =total variable costs

x = number of units produce.

Now, according to the question given for Sonam Plastic Industry:

Fixed cost (FC) = Rs. 30 Variable cost per unit = Rs. 4 now

Total variable cost = Variable cost per unit × x

Total variable cost = 4 × x = 4x

Now, using the formula of cost function:

C(x) = FC + Total variable cost

C(x) = 30 + 4x

Therefore, the cost function for Sonam Plastic Industry is:

C(x) = 30 + 4x

b. Revenue function :

A revenue function computes the total income earned by a company from selling a particular quantity of goods or services at given prices. The revenue function is crucial for understanding how sales translate into income and for making pricing and production decisions.

The revenue function $R(x)$ is calculated by the formula:

$R(x) = \text{Price per unit} \times \text{Quantity sold}$

$R(x) = 39x - 5x^2$

c. Break-even points :

The breakeven point is a fundamental concept in business and economics. It represents the level of production and sales at which a company's total revenue exactly equals its

total cost. At the breakeven point, the company makes neither a profit nor a loss. Understanding the breakeven point helps managers determine the minimum level of sales required to avoid losses and to set production targets.

For the breakeven points, the cost function C(x) must be equal to the revenue function R(x). This is because breakeven occurs when the money coming in from sales exactly covers all the costs of production.

Mathematically, we set:

$C(x) = R(x)$

Now, let us substitute the expressions we have derived for C(x) and the given expression for R(x):

$30 + 4x = 39x - 5x2$

Starting with:

$30 + 4x = 39x - 5x2$

$30 + 4x - 39x + 5x2=0$

The right-hand side becomes zero:

$5x2-35x+30=0$

Comparing this equation to the standard quadratic equation $ax2 + bx + c = 0$, we identify:

a = 5 (coefficient of x2)

b = −35 (coefficient of x)

c = 30(constant term)

To find the values of x , we can use the quadratic formula. The quadratic formula define that for any quadratic equation $ax2 + bx + c = 0$, the solutions are given by:

$x = [-b \pm \sqrt{}(b2 - 4ac)] / 2a$

First,

$x = [-(-35)\pm\sqrt{}(-35)2-4.5.30]/2.5$

Now,

$X1 = [-(-35)+\sqrt{}(-35)2-4.5.30]/2.5$

X1=6 unit

$X2= [-(-35)-\sqrt{(-35)2-4.5.30}]/2.5$

X2=1 unit

Therefore, the breakeven quantities are x = 1 unit and x = 6 units.

the revenue function R(x) = 39x − 5x2

For x1 = 6 unit :

R1 = R(x) = 39.(6) − 5(6)2

R1 =Rs. 54

For x2 = 1:

R2 = 39.(1) − 5(1)2

R2 =Rs. 34

d. Graphical representation breakeven point :

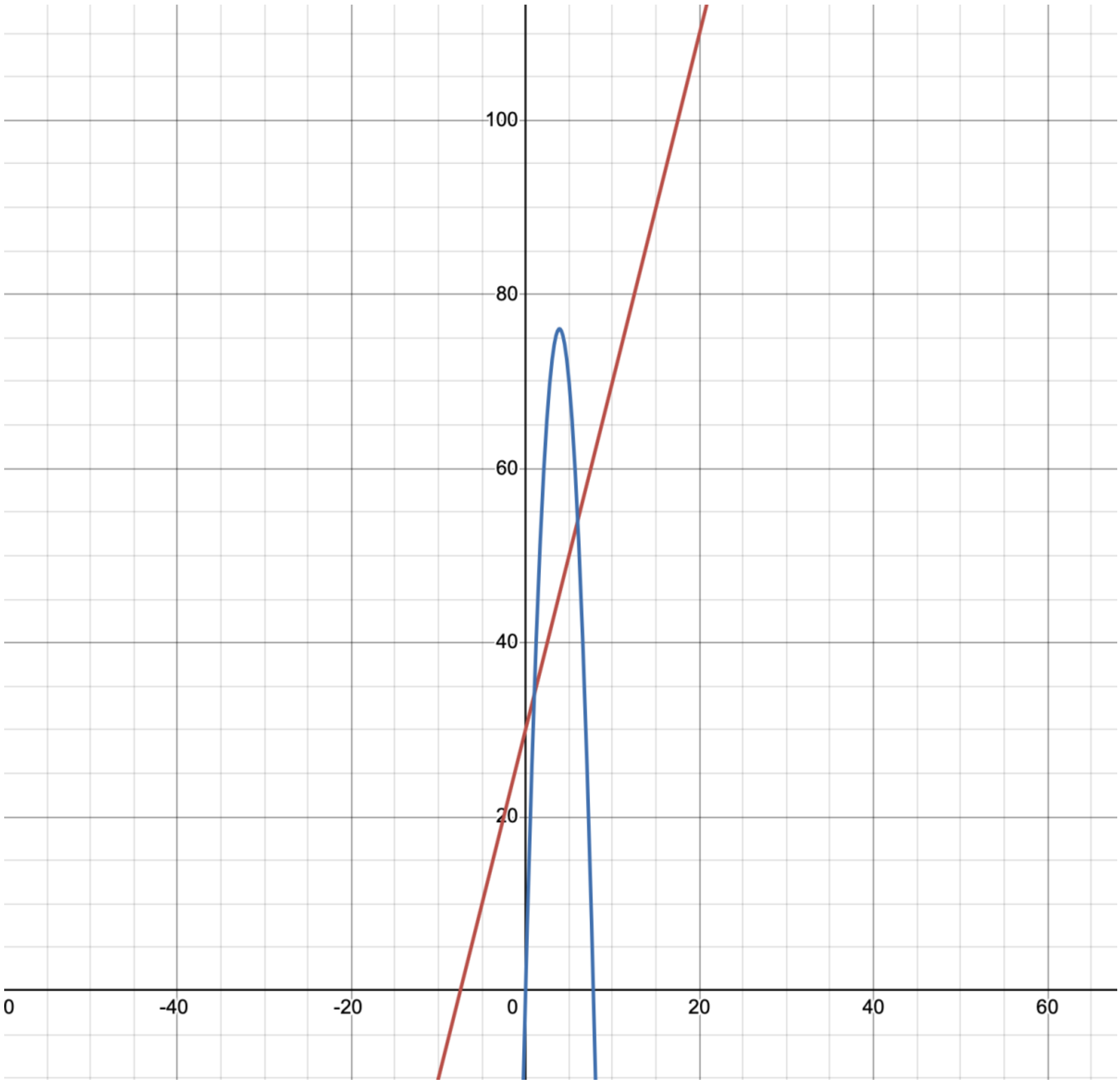


Figure 11:Graphical representation breakeven point

As from the graph, the breakeven point are (6,54) and (1,34).

e. Profit function

The profit function is one of the most important concepts in business economics. Profit is defined as the difference between total revenue and total cost. When revenue is greater than cost, the company makes a profit. When cost is greater than revenue, the company incurs a loss. The profit function allows managers to determine how profit changes with different levels of production and to identify the production level that yields the maximum possible profit.

The profit function is given by the expression:

$P(x) = R(x) - C(x)$

Where:

- $P(x)$ represents the total profit when x units are produced and sold,
- $R(x)$ represents the total revenue from selling x units,
- $C(x)$ represents the total cost of producing x units.

Now, let us substitute the expressions we have for R(x) and C(x).

$R(x) = 39x - 5x2$

$C(x) = 30 + 4x$

Using the formula of profit function:

$P(x) = R(x) - C(x)$

$P(x) = (39x - 5x2) - (30 + 4x)$

$P(x) = 39x - 5x2 - 30 - 4x$

$P(x) = -5x2 + 35x - 30$

Therefore, the profit function for Sonam Plastic Industry is:

$P(x) = -5x2 + 35x - 30$

f. Maximum profit and number of units

Now that we have the profit function, we want to find the production level that maximizes profit. This is a classic optimization problem. Since the profit function is quadratic with a negative leading coefficient, the maximum occurs at the vertex of the parabola.

The profit function we calculated is:

$P(x) = -5x2 + 35x - 30$

Comparing this equation to the standard quadratic form $ax2 + bx + c$, we can identify the coefficients:

$a = -5$ (coefficient of $x2$)

$b = 35$ (coefficient of x)

$c = -30$ (constant term)

For any quadratic function of the form $ax2 + bx + c$, the x-coordinate of the vertex (which gives either the maximum or minimum value, depending on the sign of a) is given by the formula:

$x = -b / 2a$

Let us substitute our values of b and a into this formula:

$x = -(35) / (2 \times -5)$

$x = -35 / (-10)$

When we divide a negative number by a negative number, the result is positive:

$x = 35 / 10$

$x = 3.5$

Therefore, the number of units that maximizes the profit is:

$x = 3.5$ units

Now, to find the maximum profit

$P(x) = -5x2 + 35x - 30$

$P(x) = -5 \times (3.5)2 + 35 \times 3.5 - 30$

$= 31.25$

Therefore, the maximum profit is:

$P(x) =$ Rs. 31.25

Evidences :

Evidence of Problem 2 :

Problem-2 solution using simplex Method

step1 : De-fine variables.

Let

x_1 = Number of days plants 1 operators

x_2 = Number of days plants 2 operators

x_3 = Number of day plant 3 operators.

step2 : objective function

Minimize

$$Z = 55000x_1 + 60000x_2 + 60000x_3$$

subject to:

$$8000x_1 + 6000x_2 + 12000x_3 \geq 300000$$

$$4000x_1 + 6000x_2 + 4000x_3 \geq 172000$$

$$8000x_1 + 3000x_2 + 8000x_3 \geq 249500$$

$$x_1 + x_2 + x_3 \geq 0$$

step3 : convert into standard form

Divide all eq by 1000 to simplify

$$8x_1 + 6x_2 + 12x_3 \geq 300$$

$$4x_1 + 6x_2 + 4x_3 \geq 172$$

$$8x_1 + 3x_2 + 8x_3 \geq 249.5$$

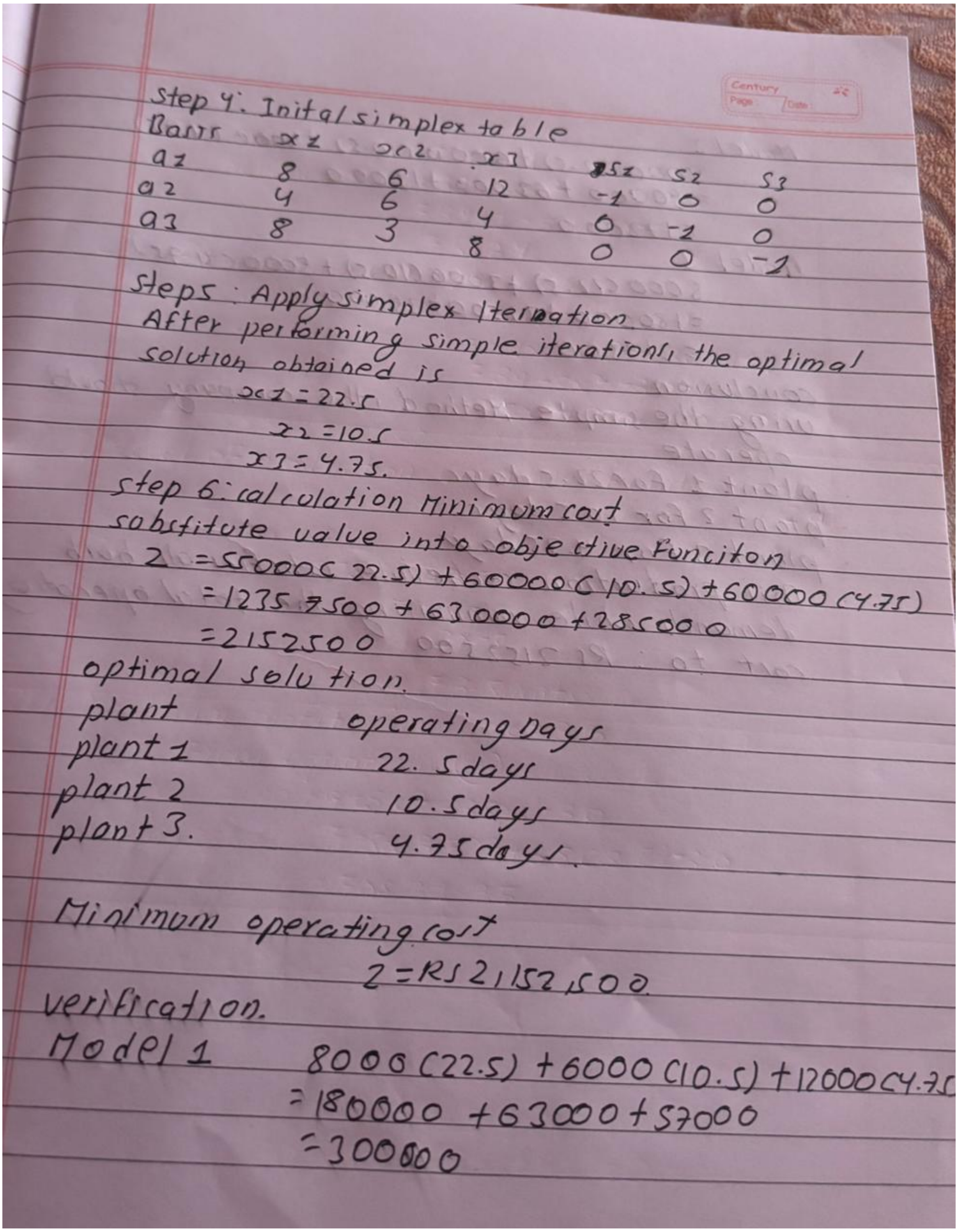
Now, introduce surplus and artifical variables:

$$8x_1 + 6x_2 + 12x_3 - S_1 + a_1 = 300$$

$$4x_1 + 6x_2 + 4x_3 - S_2 + a_2 = 172$$

$$8x_1 + 3x_2 + 8x_3 - S_3 + a_3 = 249.5.$$

Figure 12:Evidence of problem 2


Step 4: Inital simplex table

Basis	x1	x2	x3	S1	S2	S3
a1	8	6	12	-1	0	0
a2	4	6	4	0	-1	0
a3	8	3	8	0	0	-1

Steps: Apply simplex Iteration.
After performing simple iterations, the optimal solution obtained is

$x1 = 22.5$

$x2 = 10.5$

$x3 = 4.75$.

step 6: calculation Minimum cost
substitute value into objective funciton

$Z = 55000(22.5) + 60000(10.5) + 60000(4.75)$

$= 1235 7500 + 630000 + 285000$

$= 2152500$

optimal solution.

plant	operating Days
plant 1	22.5 days
plant 2	10.5 days
plant 3.	4.75 days.

Minimum operating cost

$Z = Rs\ 2,152,500.$

verification.

Model 1 $8000(22.5) + 6000(10.5) + 12000(4.75$

$= 180000 + 63000 + 57000$

$= 300000$

Figure 13:Evidence of problem 2

Model 2

$4000(22.5) + 6000(10.5) + 4000(4.75)$
$= 90000 + 63000 + 19000$
$= 172000$

Model 3

$8000(22.5) + 3000(10.5) + 8000(4.75)$
$= 180000 + 31500 + 38000$
$= 249500$

conclusion

using the simple Method, the company should operate

- plant 1 For 22.5 days
- plant 2 For 10.5 days.
- plant 3 For 4.75 days.

This production sechedule satisfies all bulb demands while minimizing the total operating cost to : Rs 2152500

Figure 14:Evidence of problem 2

Evidence of graphical representation of feasible region :

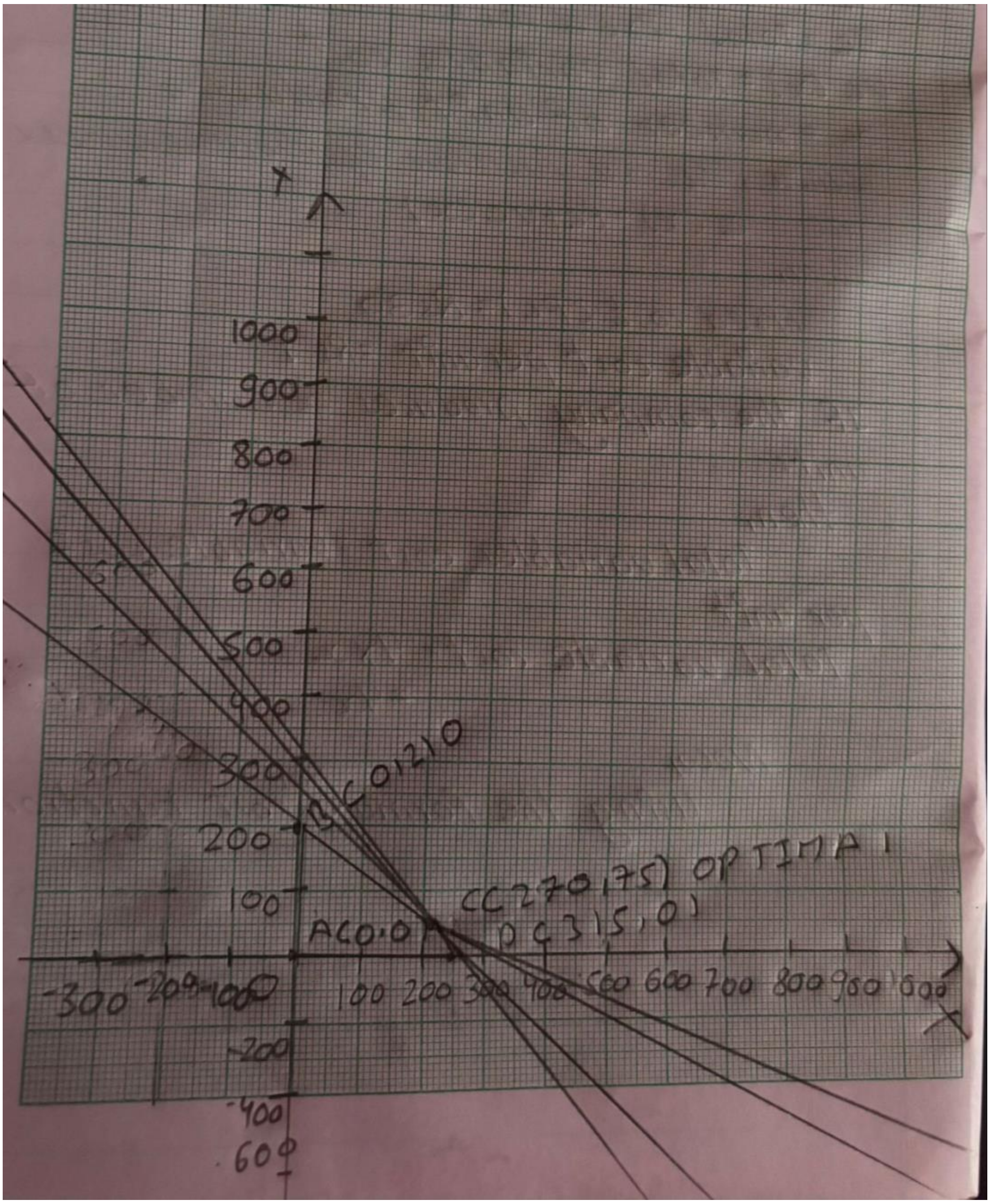


Figure 15:Evidence of graphical representation of feasible region

Evidence for Problem 3 :

Problem 3:-

Let,

$C(x)$ = total production cost

Fc = Total fixed cost

$Vc(x)$ = Total variables cost

x = number of unit produced

where,

The cost function equation is expressed as

$C(x) = Fc + Vc(x)$

Now,

Fixed cost (Fc) = Rs 30

variables cost per unit = Rs 4

If the company produces x number of units

then,

Total variables cost = variable cost per unit $\times x$

Total variables cost $= 4 \times x = 4x$.

Now, using the formula of cost function

$C(x) = Fc +$ total variable cost

$C(x) = 30 + 4x$

$\therefore$ Cost function for sonam plastic Industry

$C(x) = 30 + 4x$

$R(x) = 39 - 5x^2$

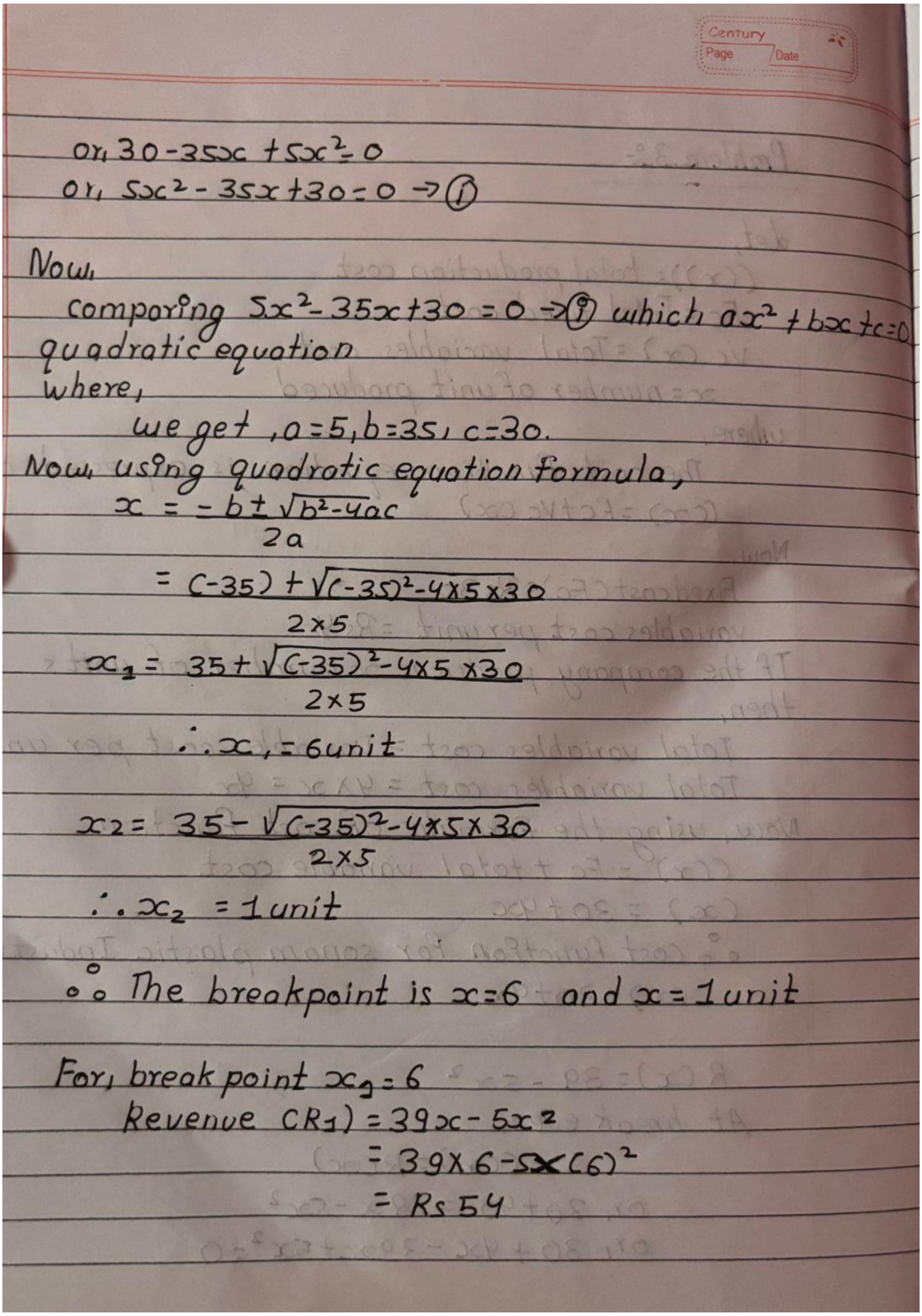
At break even,

$C(x) = R(x)$

or, $30 + 4x = 39x - 5x^2$

or, $30 + 4x - 39x + 5x^2 = 0$

Figure 16 : Evidence of problem 3



or, $30-35x+5x^2=0$
or, $5x^2-35x+30=0 \rightarrow$ (i)

Now,
comparing $5x^2-35x+30=0 \rightarrow$ (i) which $ax^2+bx+c=0$
quadratic equation
where,
we get, $a=5, b=35, c=30$.
Now using quadratic equation formula,

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{(-35)+\sqrt{(-35)^2-4\times5\times30}}{2\times5}$$

$$x_1 = \frac{35+\sqrt{(-35)^2-4\times5\times30}}{2\times5}$$

$\therefore x_1 = 6$ unit

$$x_2 = \frac{35-\sqrt{(-35)^2-4\times5\times30}}{2\times5}$$

$\therefore x_2 = 1$ unit

$\therefore$ The breakpoint is $x=6$ and $x=1$ unit

For, break point $x_1=6$
Revenue $(R_1) = 39x-5x^2$
$= 39\times6-5\times(6)^2$
$=$ Rs 54

Figure 17:Evidence of problem 3

Century
Page: Date:

For, breakever point $x_1 = 1$

Revenue $(R) = 39x - 5x^2$

$= 39 \times 1 - 5 \times (1)^2$

$= Rs\ 34$

c Profit Function $(P)(x)$

Profit Revenue $P(x) - cost(x)$

$= 39x - 5x^2 - (30 + 4x)$

$= 39 - 5x^2 - 30 - 4x$

$= 5x^2 + 35x - 30 \rightarrow (i)$

i) Level of production (number of units $x = \frac{-b}{2a}$

where,

$a = -5,\ b = 35$

$= \frac{-35}{2 \times -5}$

$= 3.5$ units

ii) Maximum profit, where $x = 3.5$

From equation (ii)

$= -5x^2 + 35x - 30$

$= -5(3.5)^2 + 35(3.5) - 30$

$= Rs\ 31.35$

Graphical representation of problem 3

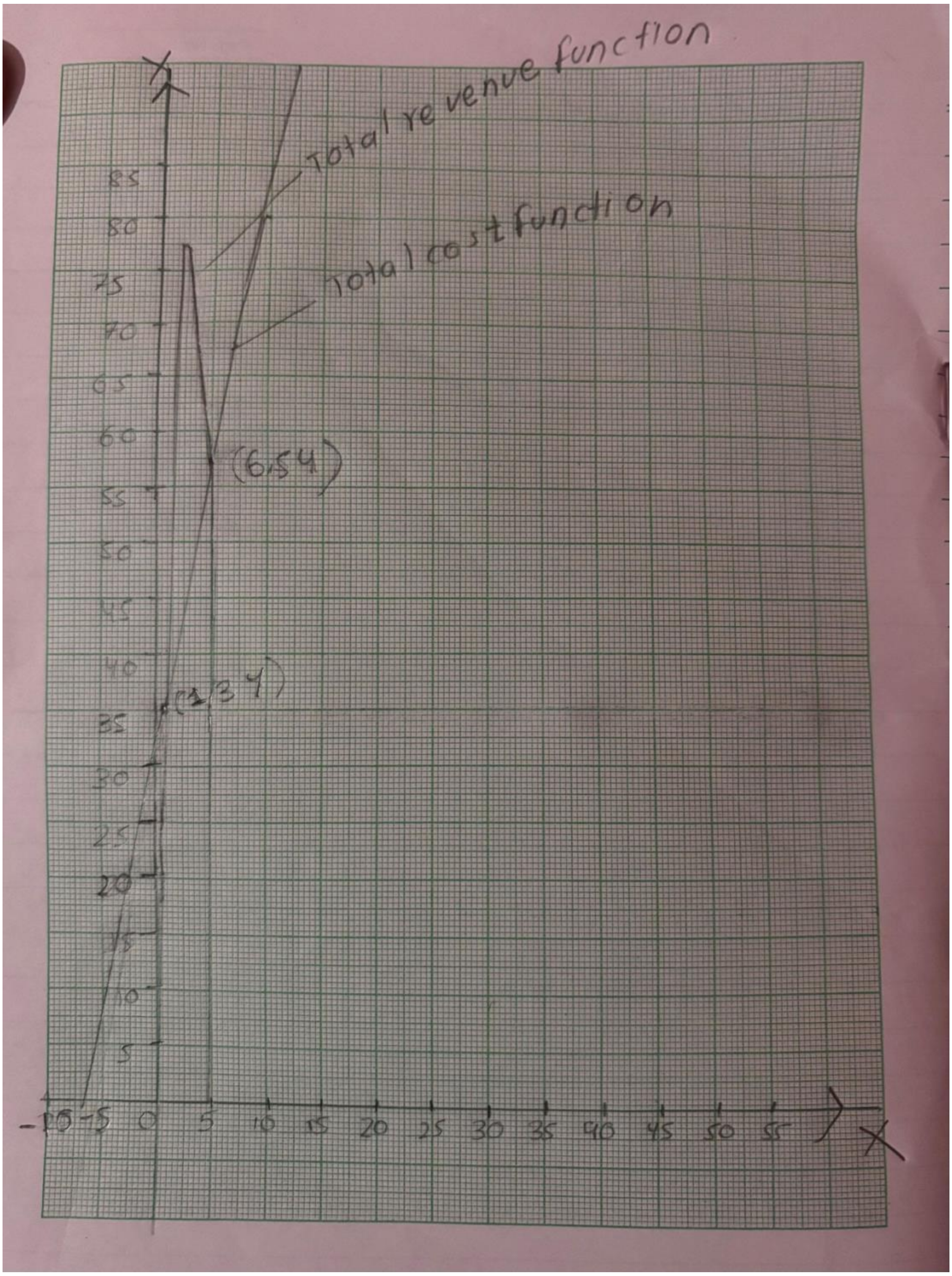


Figure 18:Graphical representation of problem 3

Appendix: Logbook Entry Sheet

Logbook Entry Sheet

Meeting No: 2 Date: 18th May

Start Time: 11:00 AM Finish Time: 1:00PM

Items Discussed: Discussed project requrements and divided tasks

Achievements: Completed project draft

Problems (if any):

Student's Name	Student's Signature
Kripa Rana	kripa rana
Shristi Paudel	shristipaudel

Appendix: Logbook Entry Sheet

Logbook Entry Sheet

Meeting No: 3 Date: 20th May

Start Time: 1:00 AM Finish Time: 3:00PM

Items Discussed: Reviewed Excel sheet formatting and data entry methods.

Achievements: Finished data for sample records.

Problems (if any): Some data entries were duplicated.

Tasks for Next Meeting: correct records and test formula

Student's Name	Student's Signature
Kripa Rana	kripa rana
Shristi Paudel	shristipaudel

Appendix: Logbook Entry Sheet

Logbook Entry Sheet

Meeting No: 4 Date: 19th May 22, 2026

Start Time: 10:00AM Finish Time: 1:00PM

Items Discussed: Discussed graphical solution for Problem 2 Part B.

Achievements: Found feasible region, corner points, and optimal value

Problems (if any):

Tasks for Next Meeting: draw final graph neatly.

Student's Name	Student's Signature
Kripa Rana	kripa rana
Shristi Paudel	shristipaudel

Appendix: Logbook Entry Sheet

Logbook Entry Sheet

Meeting No: 5 Date: 21 May,2026

Start Time: 11:00PM Finish Time: 2:00PM

Items Discussed: Discussed Problem 3 breakeven analysis and profit maximization.

Achievements: Derived total cost function, profit function, and maximum profit point.

Problems (if any):

Tasks for Next Meeting: Finalize report formatting and presentation slides

Student's Name	Student's Signature
Kripa Rana	kripa rana
Shristi Paudel	shristipaudel

Appendix: Logbook Entry Sheet

Logbook Entry Sheet

Meeting No: 1 Date: 17th May 22, 2026

Start Time: 11:00PM Finish Time: 2:00PM

Items Discussed: Discussed Problem 1 tax calculation procedure and assigned tasks to group members

Achievements: Completed tax slab analysis and yearly tax calculation formulas.

Problems (if any):

Tasks for Next Meeting: : Prepare Excel formulas and screenshots for Question 1.

Student's Name	Student's Signature
Kripa Rana	kripa rana
Shristi Paudel	shristipaudel